Activity 12 · Bounded Demand Forecast

TGS-2022017520 · v14.0 · Synthetic training data

Goal

Six synthetic monthly enquiry counts are used to forecast one month. Claude Code must compare a last-value baseline and three-month moving average, preserving seasonality and short-history uncertainty.

Before you start

Use a copied activity folder,Claude Cowork or Claude Code as specified,and a spreadsheet or Python3 for independent checks. No production credentials are needed. Product access varies;CSV fallback remains valid.

Mock data and assets

mock-data/demand.csv

All named customers,competitors,campaigns and values are fictional. Assets include brand,source,measurement and decision contracts.

Mechanism

Check regular intervals -> reserve test period -> compute naive baseline -> compare moving average -> record error -> state scenario range -> monitor change.

AI Strategy’s demand-analysis examples are adapted into baseline comparison,held-out error and campaign-change monitoring. Short-history model precision is explicitly bounded.

Worked calculation

September last-value forecast=104; three-month mean=(96+100+104)/3=100. At August, rolling mean of May-July=93.33; absolute error=10.67.

Step-by-step

1. Inspect demand.csv and identify the June campaign change.
2. Read assets/forecast-protocol.md and reserve August as holdout.
3. Ask Claude Code Prompt 1 to implement baseline and moving average.
4. Review that August is not used in its own predicted value.
5. Run the analysis or supplied validate.py.
6. Verify September 104 and 100; August MA error 10.67.
7. Write an assumption-based low/base/high scenario distinct from a statistical interval.
8. Save forecast CSV,script and assumptions note under outputs.

Prompt1

Forecast one month from six synthetic observations. Compare last-value and three-month averages. Hold August out for evaluation before forecasting September. Do not fabricate confidence intervals. Report assumptions, errors and a monitoring trigger.

Prompt2

Challenge the previous result. Find denominator errors,unsupported claims,bias,privacy risks and unexecuted tool boundaries. Cite exact row IDs or fields. Revise the report without inventing data.

Acceptance evidence

Naive 104; moving-average 100; August moving-average error 10.67; campaign shift and seasonality caveats.

Failure and recovery

The agent fits an elaborate model to six points and reports a precise probability interval. The data do not support such precision.

Use simple baselines, avoid training on the held-out month, label scenario ranges as assumptions, and discuss campaign shifts plus missing annual seasonality.

Submit

forecast.csv and forecast-assumptions.md

Save outputs in this working copy. Keep an execution note stating which product actions actually ran and which used fallback.